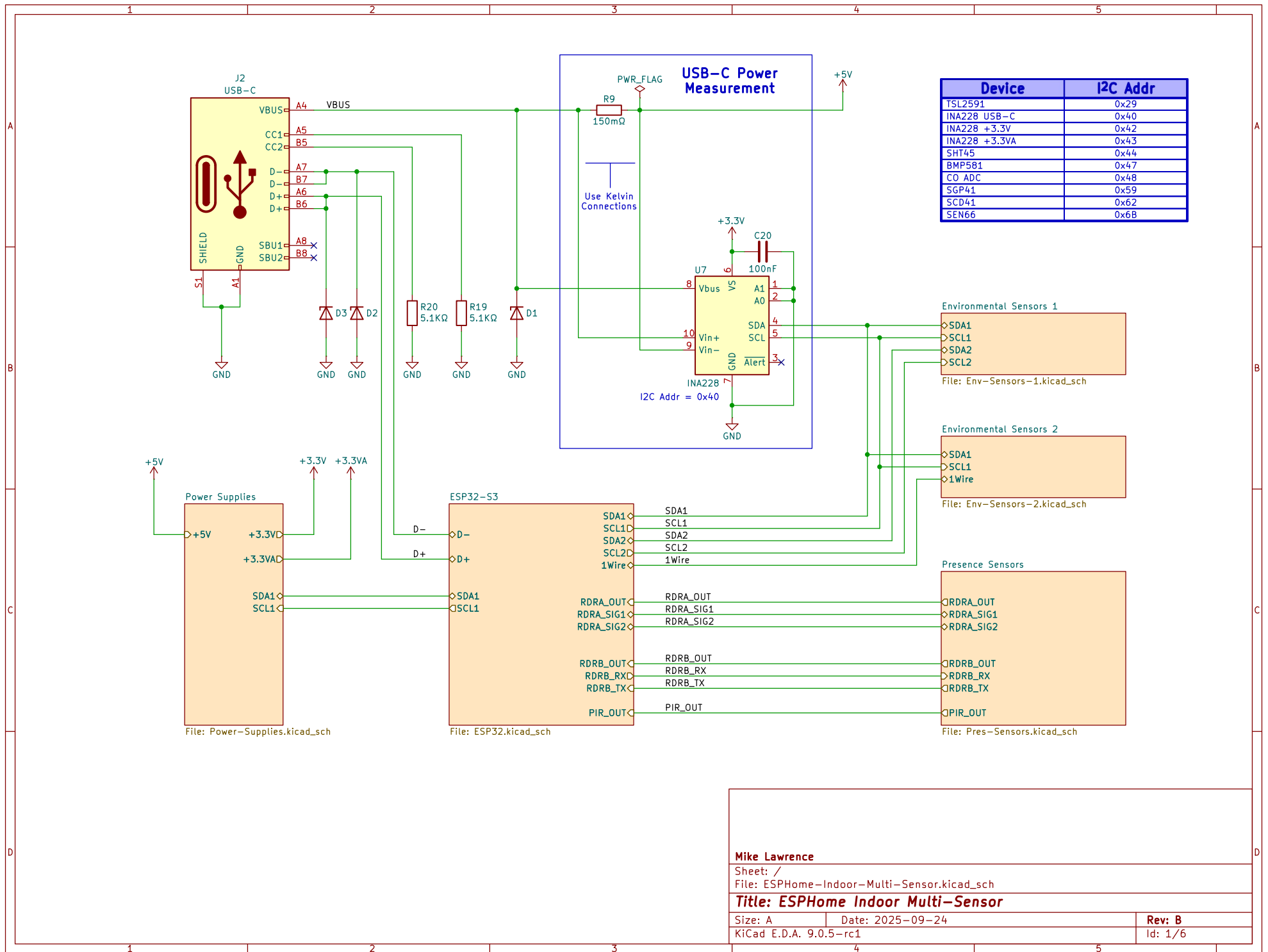




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Sheet: /

File: ESPHome-Indoor-Multi-Sensor.kicad_sch

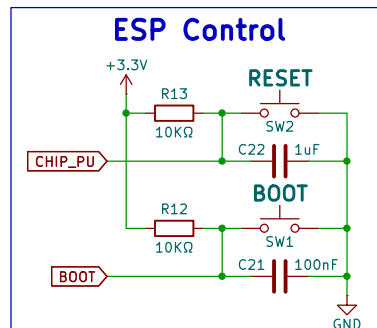
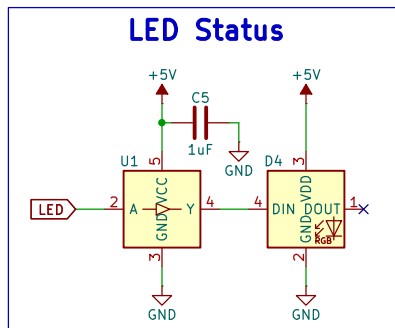
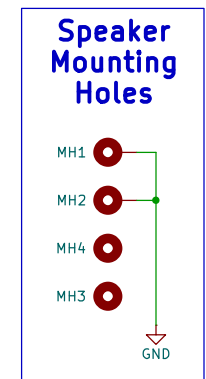
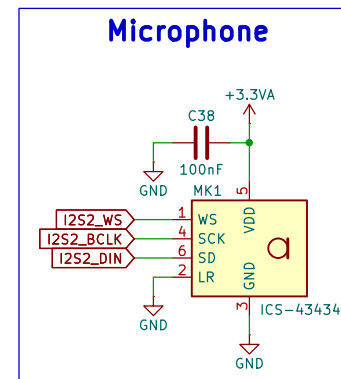
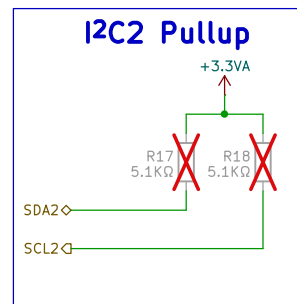
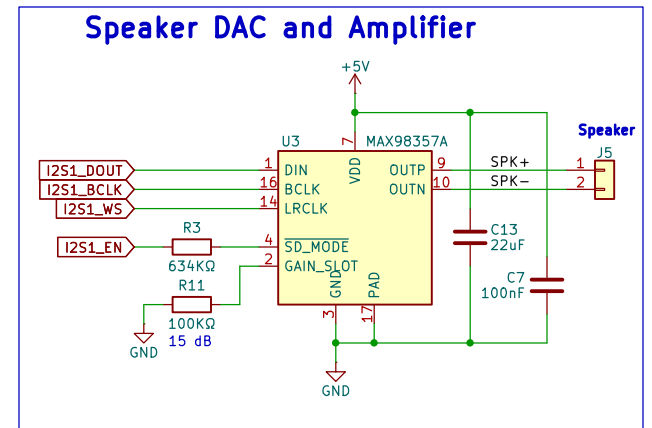
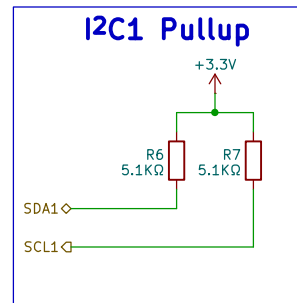
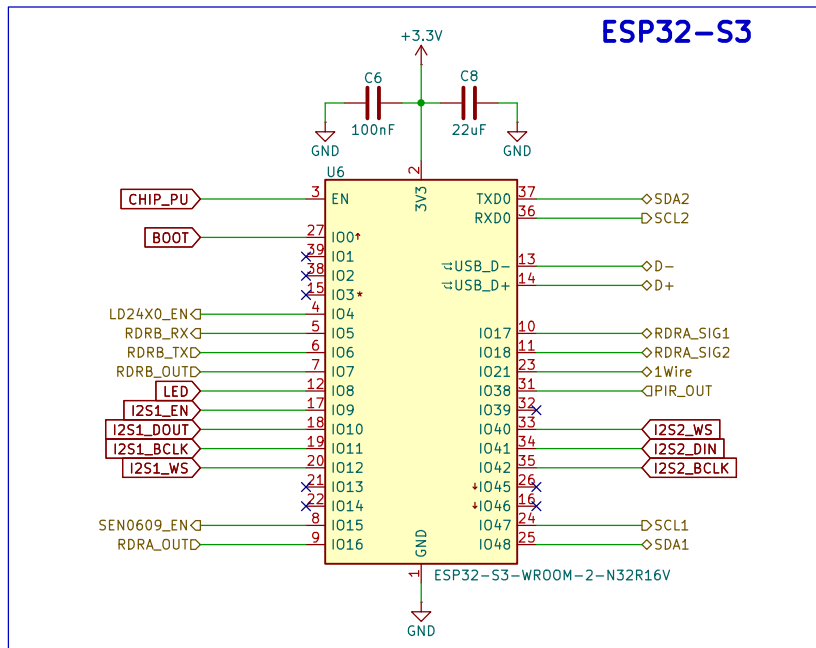
Title: ESPHome Indoor Multi-Sensor

Size: A | Date: 2025-09-24

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Rev: B

Id: 1/6



Notes

1. ESP32-S3 Strapping Pins: GPIO0, GPIO3, GPIO45, GPIO46.
2. MAX98357A DAC when enabled is configured for stereo mode.
3. SEN0609_EN and LD24X0_EN are no longer used. In order to maintain backwards compatibility these pins should not be used.
4. I2C1 pullup is 3.4K with ESP32-s 10K pullup in parallel.
5. I2C2 pullup is 10K due to ESP32-s 10K pullup.

Rev B: Support for LD2410S/LD2420 and temp for CO sensor.
Rev A: Fixed Rev - mistakes.

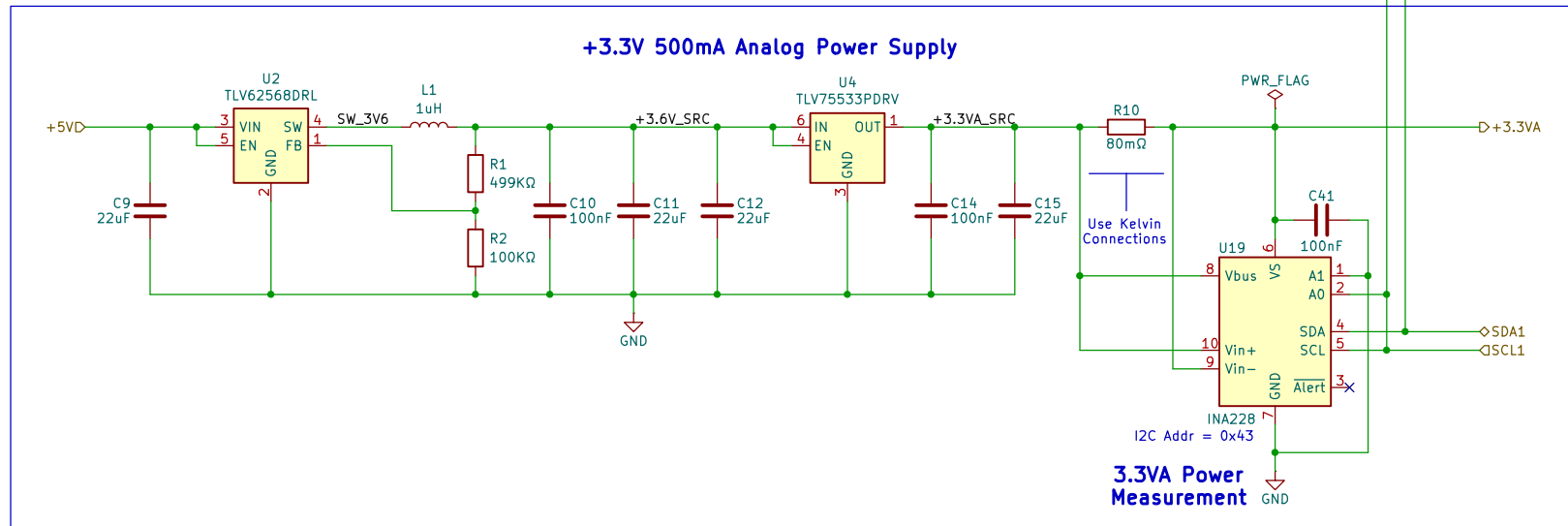
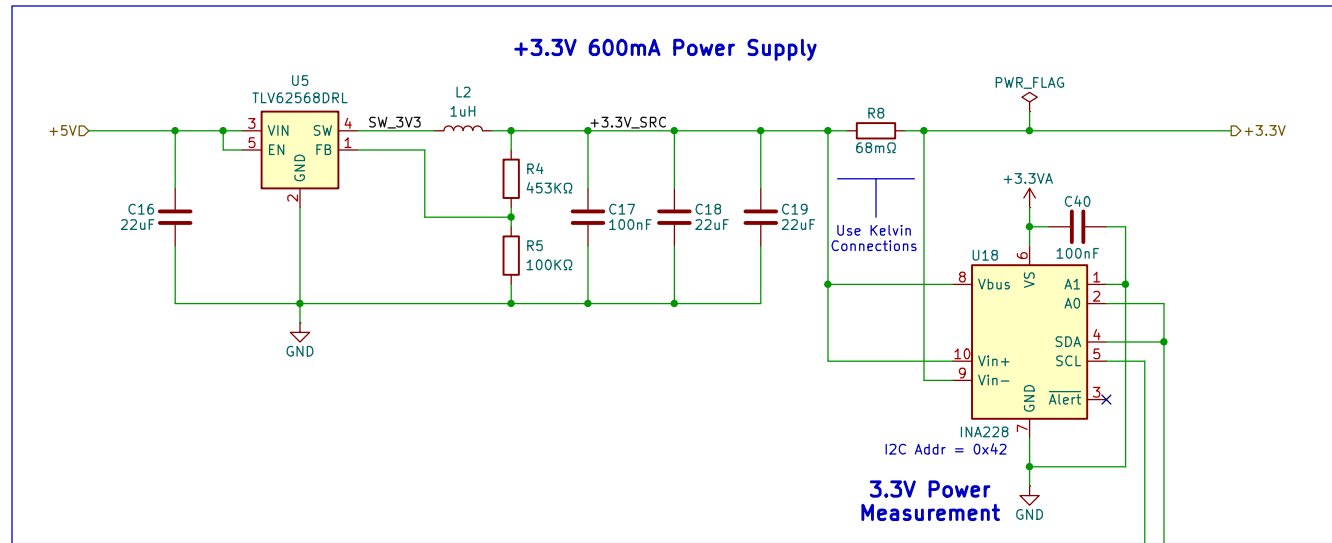
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Sheet: /ESP32-S3/
File: ESP32.kicad_sch

Title: ESPHome Indoor Multi-Sensor

Size: A Date: 2025-09-24
KiCad E.D.A. 9.0.5-rc1

Rev: B
Id: 2/6



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Sheet: /Power Supplies/

File: Power-Supplies.kicad_sch

Title: ESPHome Indoor Multi-Sensor

Size: A Date: 2025-09-24

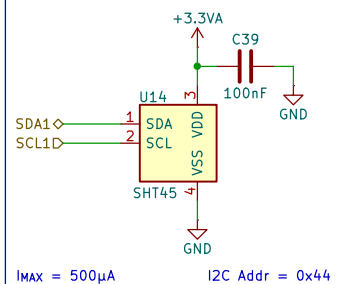
KiCad E.D.A. 9.0.5-rc1

Rev: B

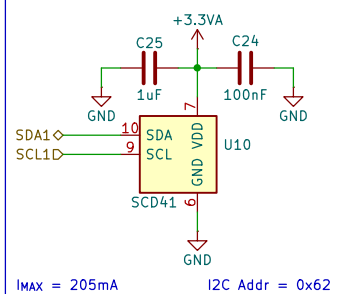
Id: 3/6

Bath or Kitchen Set

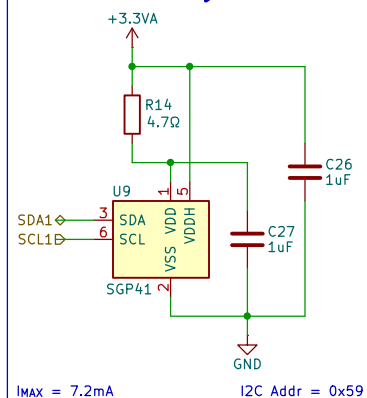
Temperature Humidity Sensor



CO2 Sensor

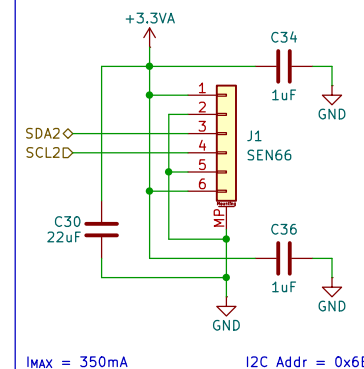


Air Quality Sensor



All-In-One Sensor

Environmental Sensor



Notes

1. Recommend one of two configurations.
 - A. SEN66.
 - B. SHT45, SCD41 and SGP41 for Bath or Kitchen.
2. Temperature & Humidity Sensor can be SHT40, SHT41 or SHT45 for increasing accuracy.
3. CO2 (SCD40 or SCD41) measures CO2, Temperature and Humidity.
4. Air Quality Sensor (SGP40 or SGP41) measures VOC, NOx. These sensors are self-heating and not recommended for measuring room temperature.
5. SEN66 measures Temperature, Humidity, CO2, VOC, NOx and PM.

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Sheet: /Environmental Sensors 1/
File: Env-Sensors-1.kicad_sch

Title: ESPHome Indoor Multi-Sensor

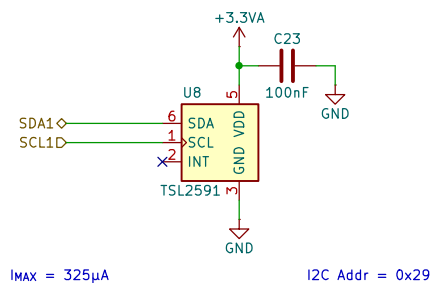
Size: A Date: 2025-09-24

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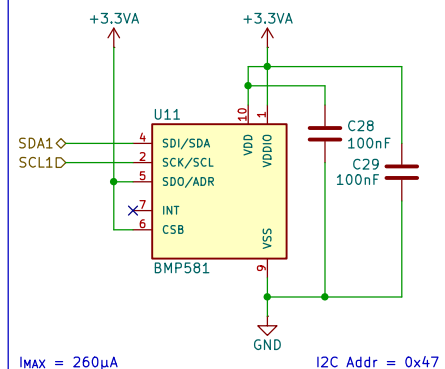
Rev: B

Id: 4/6

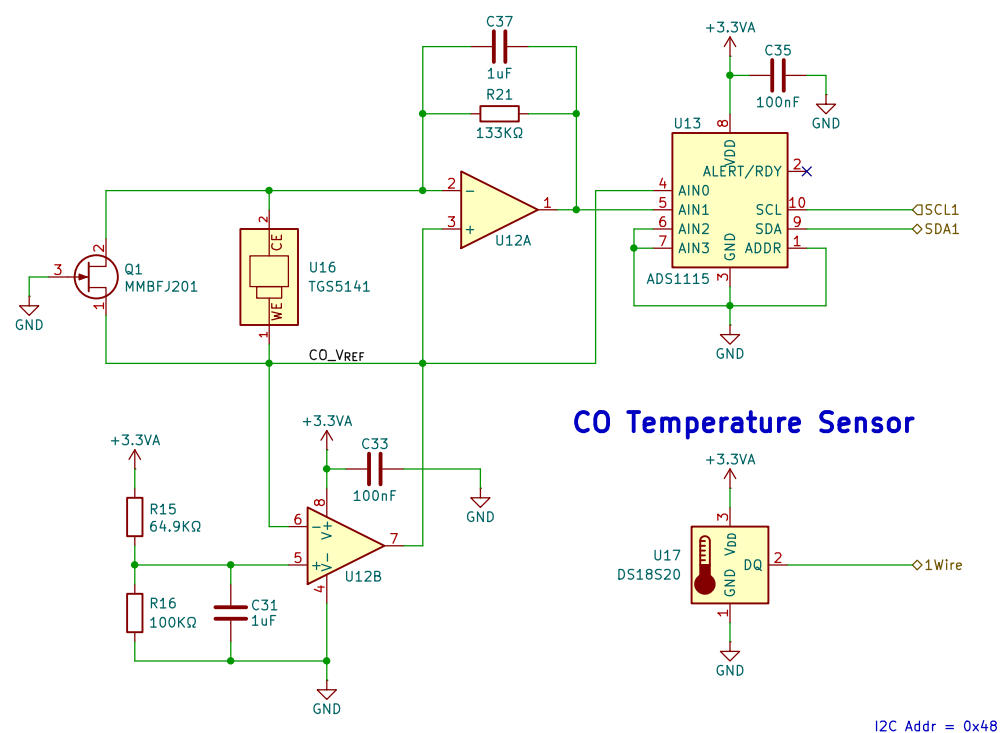
Ambient Light Sensor



Pressure Sensor



CO Temperature Sensor



Notes

1. Accuracy of CO_VREF not critical.
Needs to be around 2V to turn off Q1.
2. ADC range 0.512V.
3. Max 1000PPM.
4. Gain = $0.512V / (3.2nA \times 1.225 \times 1000PPM) = 133.333k$

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Sheet: /Environmental Sensors 2/
File: Env-Sensors-2.kicad_sch

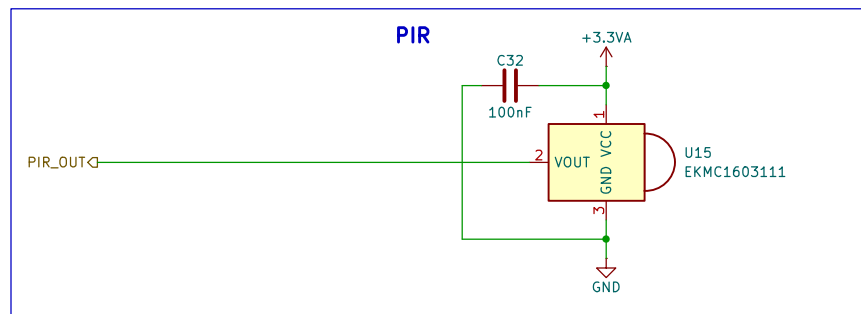
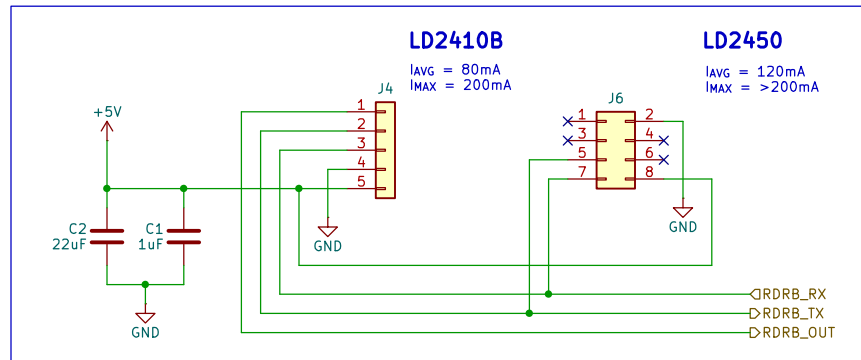
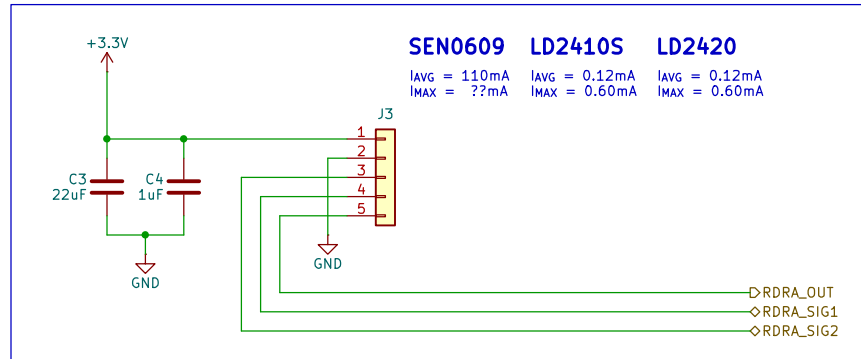
Title: ESPHome Indoor Multi-Sensor

Size: A Date: 2025-09-24

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Rev: B

Id: 5/6



Notes

1. RDRA_SIG1 is SEN0609 TX, and LD2410S/LD2420 RX.
2. RDRA_SIG2 is SEN0609 RX, and LD2410S/LD2420 TX.

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Sheet: /Presence Sensors/
 File: Pres-Sensors.kicad_sch

Title: ESPHome Indoor Multi-Sensor

Size: A Date: 2025-09-24
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Rev: B
 Id: 6/6